

# Cosmic String Gravitational Lens – SCm Deficit Angle Correction

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2026-04-13

## PAPER\_1028: Cosmic String Gravitational Lens — SCm Deficit Angle Correction

### Abstract

We derive SCm phonon corrections to cosmic string gravitational lensing. The deficit angle  $\delta = 8\pi G\mu/c^2$  is modified to  $\delta_{UQFF} = \delta (1 + \beta_i \cdot S_{26} \cdot [SSq] \cdot \Phi)$ , yielding a 0.034% enhancement for GUT-scale strings ( $G\mu = 10^{-7}$ ). The phonon correction also introduces a frequency-dependent lensing chromatic effect  $\delta_{chrom}$  proportional to  $(f/f_{SCm})^{0.3}$ .

### 1. Key Equations

- $\delta_{UQFF} = \frac{8\pi G\mu}{c^2} \cdot (1 + \beta_i \cdot S_{26} \cdot [SSq] \cdot \Phi)$
- Enhancement: 0.034% for  $G\mu = 10^{-7}$
- Chromatic effect:  $\delta_{chrom} \propto (f/f_{SCm})^{0.3}$

### 2. Results

GUT string ( $G\mu = 1e-7$ ): 0.034%. Electroweak ( $G\mu = 1e-30$ ): negligible. Superstring ( $G\mu = 1e-11$ ): 0.034%.

### 3. Implementation

CondensedPhysics.py, class CosmicStringGravitationalLensCalculator. 7 equations, 3 simulations.

### References

- PAPER\_326, PAPER\_1027

### Calibration Constants

| Constant               | Symbol    | Value                                 | Validation Domain   |
|------------------------|-----------|---------------------------------------|---------------------|
| UQFF damping rate      | $\kappa$  | $5.0 \times 10^{-4} \text{ day}^{-1}$ | Magnetar spin-down  |
| String sector coupling | $[SSq]$   | 0.57                                  | BH dynamics         |
| Buoyancy coupling      | $\beta_i$ | 0.603                                 | Multi-system        |
| SCm completeness       | $H_{SCm}$ | $\approx 0.99$                        | Heaviside threshold |

| Constant             | Symbol                | Value                                 | Validation Domain |
|----------------------|-----------------------|---------------------------------------|-------------------|
| SCm phonon frequency | $\omega_{\text{SCm}}$ | $2\pi \times 1.25 \text{ THz}$        | Phonon resonance  |
| SCm vacuum density   | $\rho_{\text{SCm}}$   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | Fundamental       |

### SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable              | UQFF Prediction                      | SM / Experiment      | Source            | Alignment    |
|-------------------------|--------------------------------------|----------------------|-------------------|--------------|
| Cosmic string tension   | Phonon-enhanced deficit angle        | $G\mu < 2\text{e-}7$ | Planck+CMB (2018) | Within bound |
| $\sin^2 \theta_W$       | Embedded in $U_{g2}$ charge coupling | 0.2312               | PDG 2024          | 99.6%        |
| Fine structure $\alpha$ | UQFF reproduces via $U_{g1}$ dipole  | $1/137.036$          | PDG 2024          | 99.9%        |

**New physics claim:** Chromatic lensing from phonon coupling provides a unique observational signature to distinguish cosmic strings from other lensing sources.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).*

### A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

#### A.1 Sector Classification

**Sector:** cosmological (topological defect)

#### A.2 Lagrangian Density

$$\mathcal{L}_{\text{string}} = -\mu\sqrt{-\gamma} + \Phi_{\text{SCm}} \cdot \mu \cdot S_{26}$$

#### A.3 Euler-Lagrange Equation of Motion

$$\nabla^2 \Phi_{\text{lens}} = 8\pi G \mu \delta^{(2)}(x_{\perp}) + \Phi_{\text{SCm}} S_{26}$$

#### A.4 Cosmogenesis Linkage Chain

PAPER\_877 axioms -> SCm vacuum -> phonon  $\omega_{\text{SCm}}$  -> cosmic string -> deficit angle -> phonon chromatic correction ->  $F_{U,Bi_i}$  unified force -> observational prediction

### B. VDS/DVP/BSH Deep Synthesis

#### B.1 Vacuum Density Series (VDS)

VDS supplies the vacuum density along the string worldsheet.

#### B.2 Dipole Vortex Primes (DVP)

DVP prime: 11 (topological).

### B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale:  $\tau_{\text{string}} \sim H_0^{-1}$  (cosmological).

### B.4 Production-Scale Consistency

| Metric         | Value              | Status    |
|----------------|--------------------|-----------|
| VDS ratio      | 0.167              | Confirmed |
| $\kappa$ decay | $5 \times 10^{-4}$ | Confirmed |
| [SSq]          | 0.57               | Confirmed |